PREVIEW



CAMBRIDGE O LEVEL

FORMULA SHEET

Mathematics (Syllabus D) 4024

Every formula you need for Paper 1 & Paper 2 · 2025–2027 syllabus

Preview

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given = printed on the exam's List of formulas Tip = exam technique

1 Number

TYPES OF NUMBER

- **Prime**: exactly two factors (1 and itself); 1 is **not** prime.
- **Rational**: can be written as $\frac{a}{b}$ (integers, $b \neq 0$), e.g. $0.\dot{3} = \frac{1}{3}$. **Irrational**: e.g. π, $\sqrt{2}$.
- **Reciprocal** of x is $\frac{1}{x}$. Squares to $15^2 = 225$; cubes $1, 8, 27, 64, 125$ and $10^3 = 1000$.

HCF AND LCM FROM PRIME FACTORS

Write each number as a product of primes, e.g. $72 = 2^3 \times 3^2$, $60 = 2^2 \times 3 \times 5$.

HCF: multiply the *common* primes using the *lowest* powers: $2^2 \times 3 = 12$.

LCM: multiply *all* primes using the *highest* powers: $2^3 \times 3^2 \times 5 = 360$.

SETS

Elements	$x \in A$ (is an element of), $x \notin A$
Number of elements	$n(A)$
Complement / empty	A' (not in A), $\emptyset$ (empty set)
Universal set	$\mathcal{E}$
Subset	$A \subseteq B$, $A \not\subseteq B$
Union	$A \cup B$ (in A or B or both)
Intersection	$A \cap B$ (in both A and B)
Two sets	$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

INDICES

Multiply	$a^m \times a^n = a^{m+n}$
Divide	$a^m \div a^n = a^{m-n}$
Power of a power	$(a^m)^n = a^{mn}$
Zero / negative	$a^0 = 1$ $\quad a^{-n} = \frac{1}{a^n}$
Fractional	$a^{\frac{1}{n}} = \sqrt[n]{a}$ $\quad a^{\frac{m}{n}} = (\sqrt[n]{a})^m$
Example	$8^{-\frac{2}{3}} = \frac{1}{(\sqrt[3]{8})^2} = \frac{1}{4}$

STANDARD FORM

Form	$A \times 10^n$
	$1 \leq A < 10$, n an integer

SURDS

Rules	$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$ $\quad \frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$
Simplify	$\sqrt{200} = \sqrt{100 \times 2} = 10\sqrt{2}$
Rationalise	$\frac{k}{\sqrt{a}} = \frac{k\sqrt{a}}{a}$
Conjugate	$\frac{k}{a + \sqrt{b}} = \frac{k(a - \sqrt{b})}{a^2 - b}$

BOUNDS (LIMITS OF ACCURACY)

A value rounded to the nearest unit u lies within $\pm\frac{1}{2}u$: e.g. 7.4 to 1 d.p. gives $7.35 \leq x < 7.45$.

UB = upper bound, **LB** = lower bound. To make an answer as big as possible, choose the bound of each value that pushes it up.

For $a + b$	UB $= a_{\text{UB}} + b_{\text{UB}}$	LB $= a_{\text{LB}} + b_{\text{LB}}$
For $a - b$	UB $= a_{\text{UB}} - b_{\text{LB}}$	LB $= a_{\text{LB}} - b_{\text{UB}}$
For $a \times b$	UB $= a_{\text{UB}} \times b_{\text{UB}}$	LB $= a_{\text{LB}} \times b_{\text{LB}}$
For $a \div b$	UB $= a_{\text{UB}} \div b_{\text{LB}}$	LB $= a_{\text{LB}} \div b_{\text{UB}}$

RATIO, RATES AND PERCENTAGES

Share in ratio $a : b$	$\frac{a}{a+b} \times$ total and $\frac{b}{a+b} \times$ total
Speed	$\text{speed} = \frac{\text{distance}}{\text{time}}$
Density	$\text{density} = \frac{\text{mass}}{\text{volume}}$
Pressure	$\text{pressure} = \frac{\text{force}}{\text{area}}$
Percentage change	$\frac{\text{new} - \text{original}}{\text{original}} \times 100\%$
Increase by $r\%$	$\times\left(1 + \frac{r}{100}\right)$
Decrease by $r\%$	$\times\left(1 - \frac{r}{100}\right)$
Percentage profit	$\frac{\text{profit}}{\text{cost price}} \times 100\%$
Reverse percentage	$\text{original} = \frac{\text{new amount}}{\text{multiplier}}$
Simple interest	$I = \frac{PRT}{100}$
Compound interest	$A = P\left(1 + \frac{r}{100}\right)^n$
Depreciation / decay	$A = P\left(1 - \frac{r}{100}\right)^n$

> **Tip** Reverse percentages: a price *after* a 20% rise is 120% of the original, so divide by 1.2 — never take 20% off the new price. Compound interest earned $= A - P$.

TIME

1 hour $= 60$ min; 3.25 hours $= 3$ h 15 min (not 3 h 25 min). 1 day $= 24$ h; 1 year $= 365$ days.

RECURRING DECIMALS TO FRACTIONS

$x = 0.2\dot{7}\dot{7}$: $100x = 27.2\dot{7}\dot{7}$, so $99x = 27$ and $x = \frac{27}{99} = \frac{3}{11}$.

2 Algebra & graphs

EXPANDING AND FACTORISING

Perfect squares	$(a + b)^2 = a^2 + 2ab + b^2$
	$(a - b)^2 = a^2 - 2ab + b^2$
Difference of two squares	$a^2 - b^2 = (a + b)(a - b)$
Grouping	$ax + ay + bx + by = (a + b)(x + y)$

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